

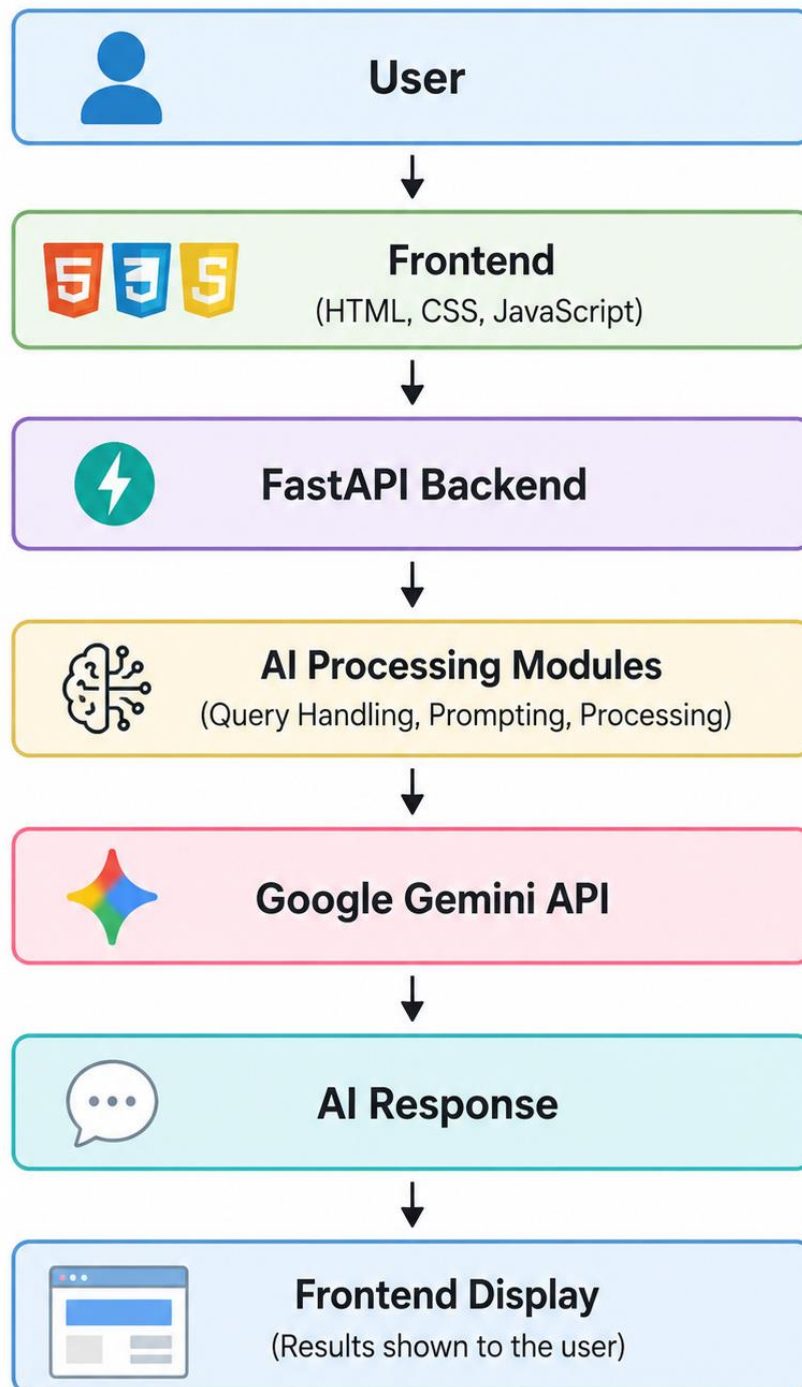
Design Document

Introduction

Explain the purpose of the project and its architecture.

System Architecture

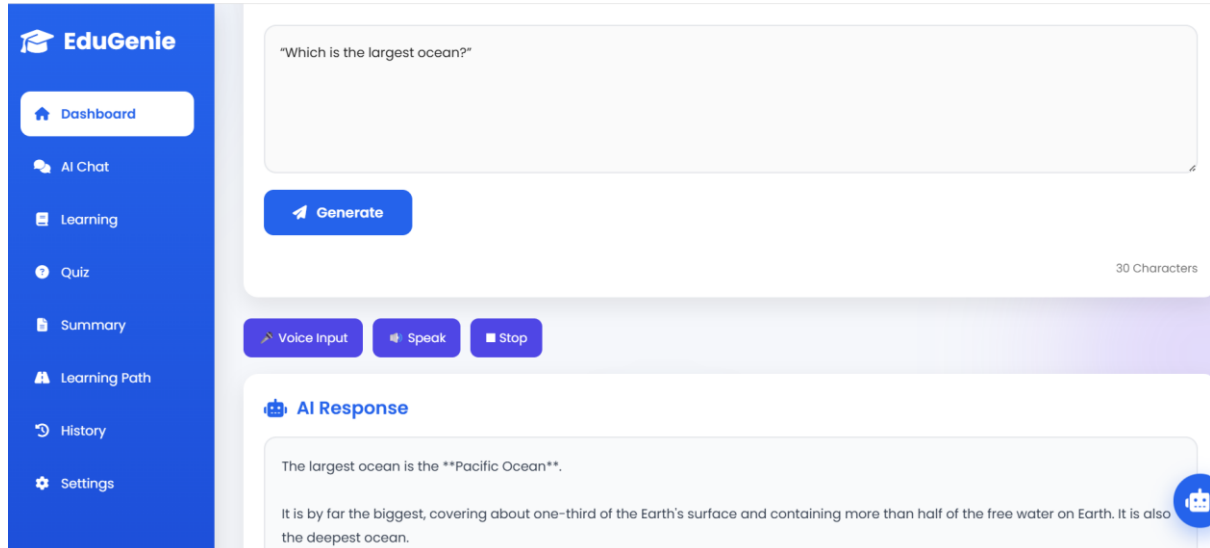
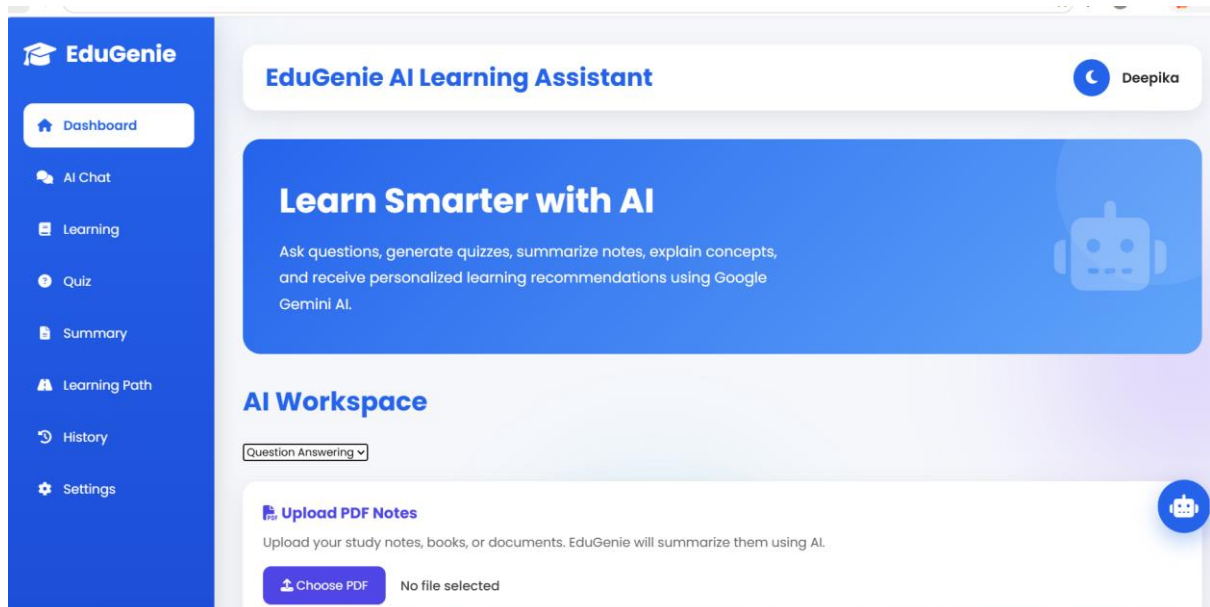
Describe how the frontend, backend, AI modules, and Gemini API interact.



MODULES:

Question Answering

This module allows users to ask educational questions and receive AI-generated responses using Google Gemini.



Concept Explanation

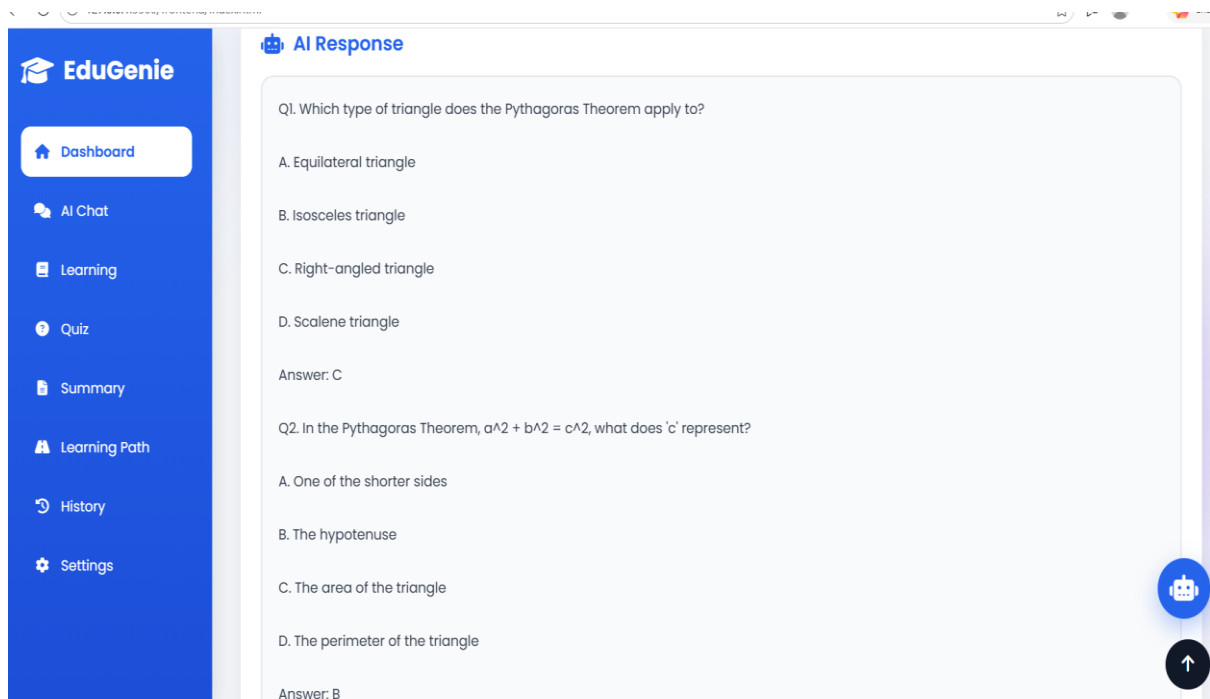
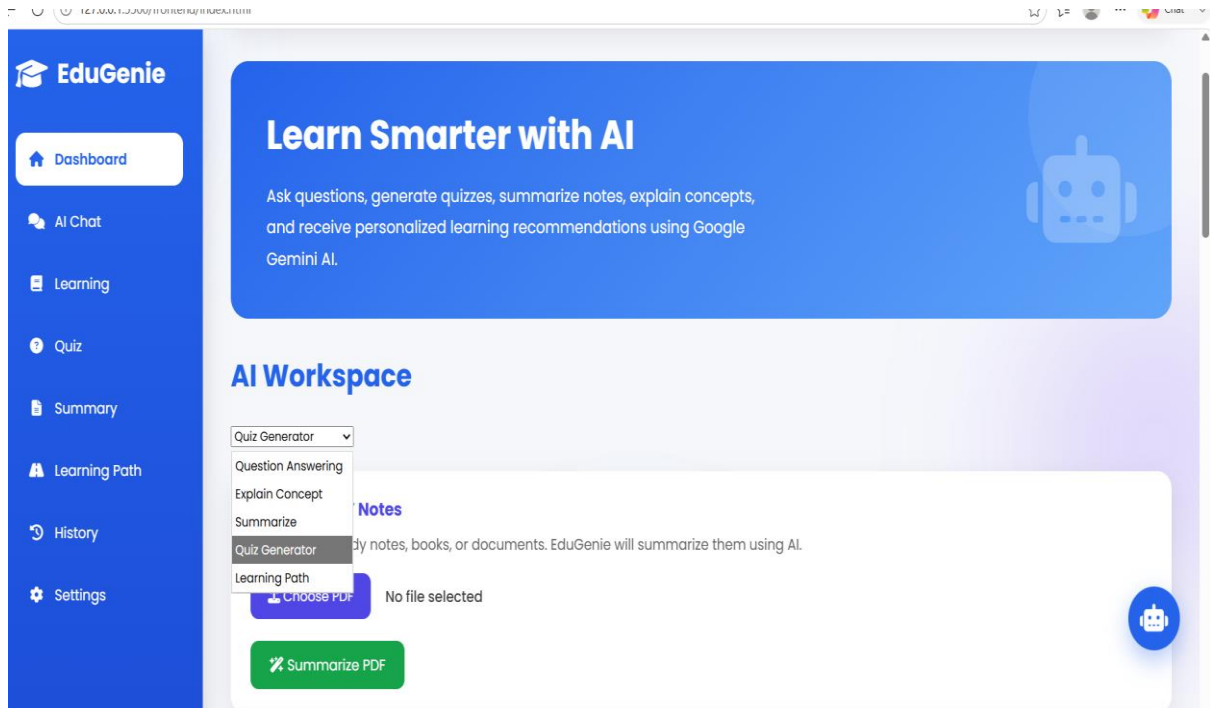
This module explains difficult concepts in simple and understandable language suitable for students.

Text Summarization

Users can paste lengthy notes or articles and obtain concise summaries.

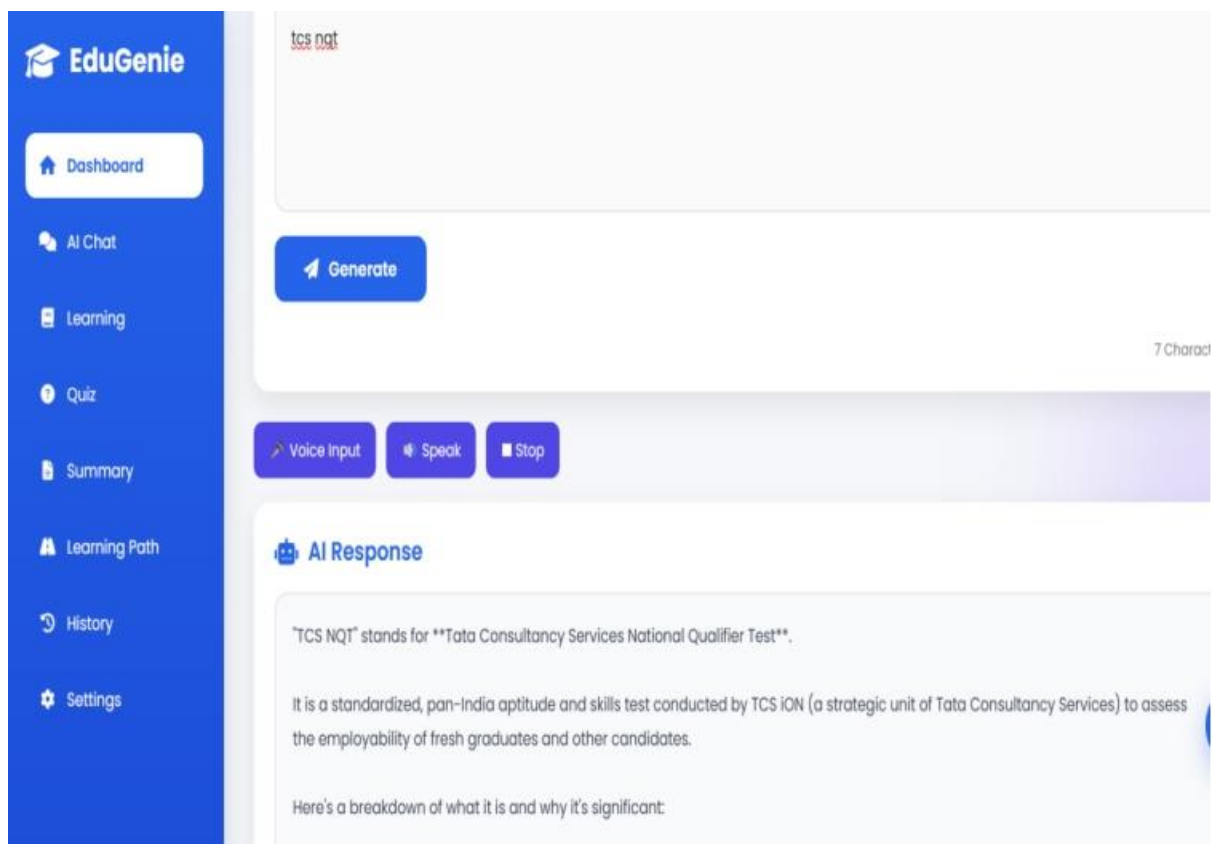
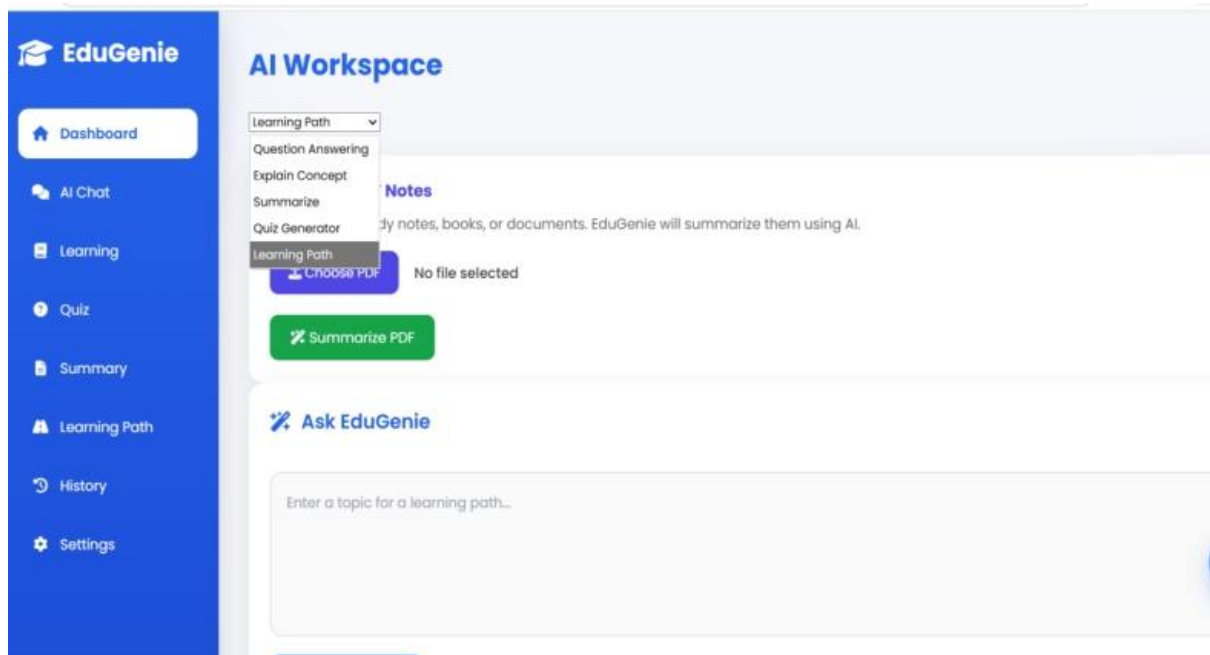
Quiz Generator

Automatically creates multiple-choice questions based on user-provided topics.



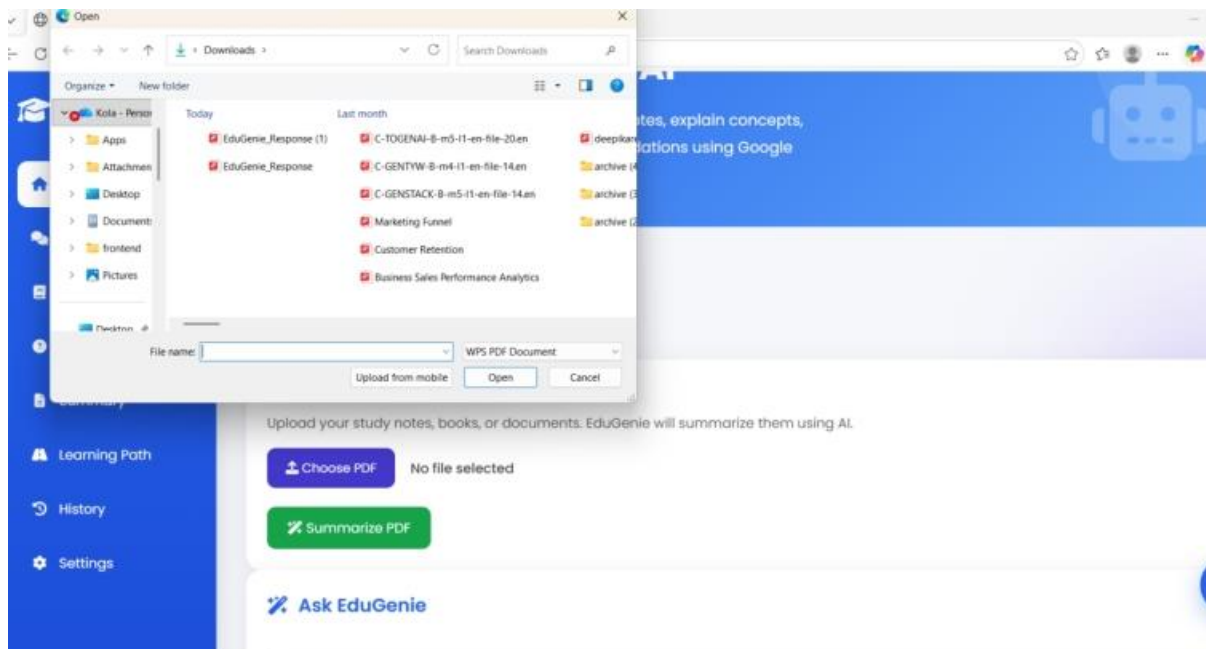
Learning Path Recommendation

Suggests personalized learning paths based on the selected topic.



PDF Upload & Summarization

Allows users to upload PDF documents. The system extracts text and generates concise summaries using AI.



Voice Input

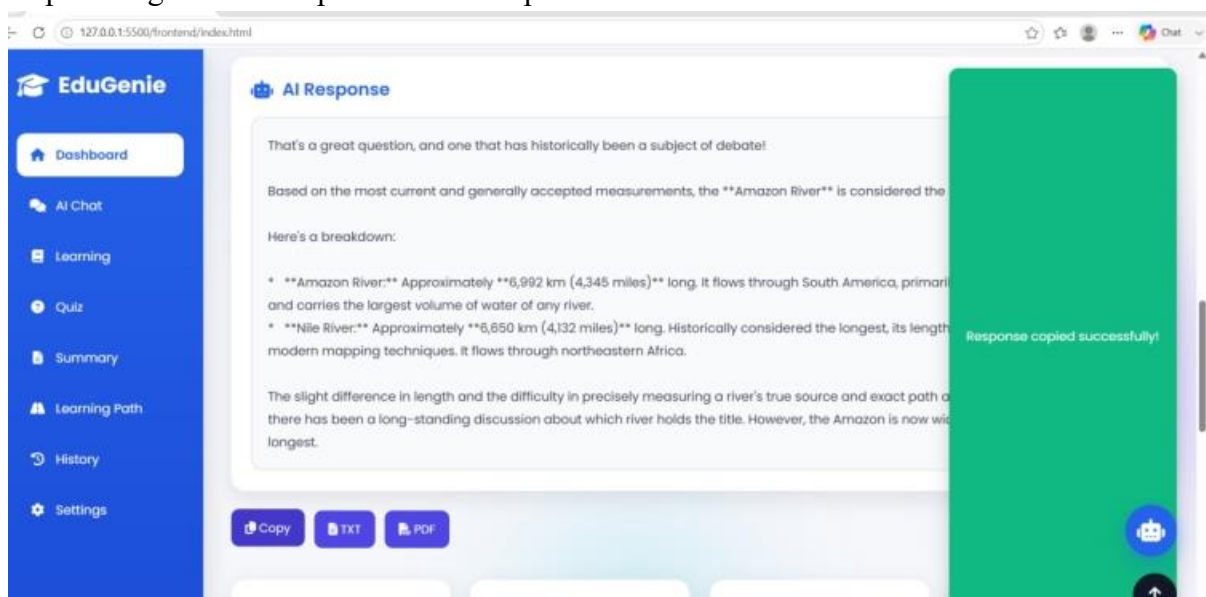
Users can ask questions using voice through the browser's Speech Recognition API.

Text-to-Speech

Reads AI-generated responses aloud to improve accessibility.

Copy Response

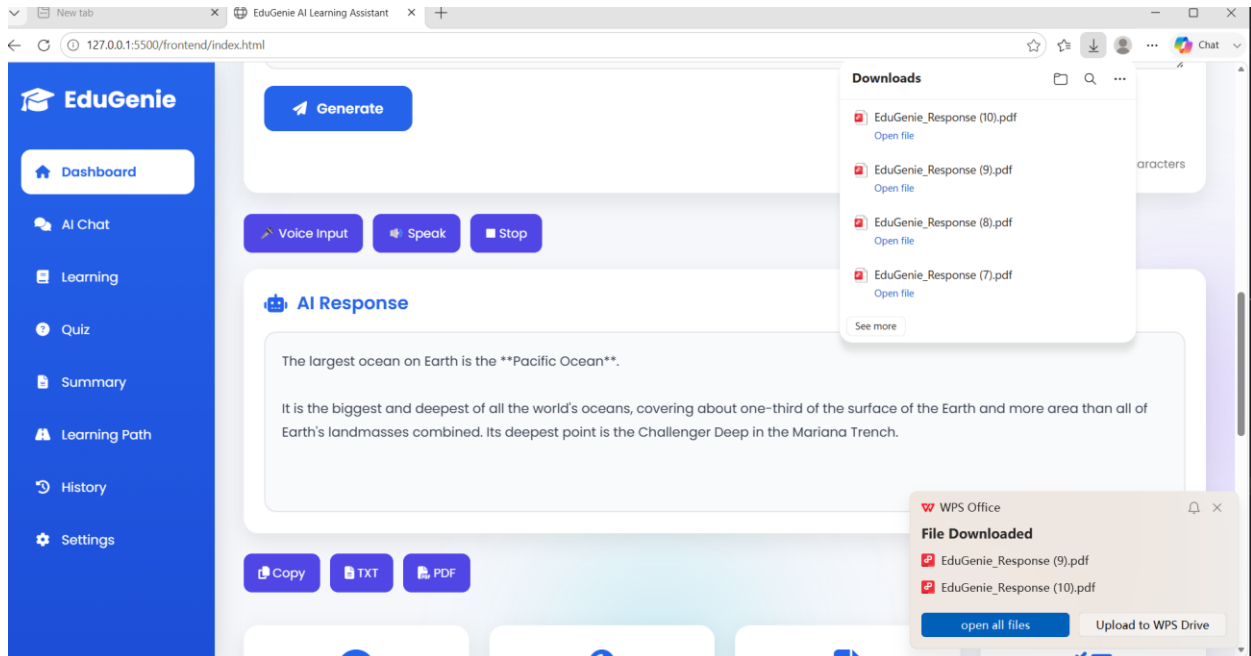
Copies AI-generated responses to the clipboard with one click.



Download Response

Users can download AI responses in PDF or TXT format.

- AI-powered educational support



Advantages

- Modular architecture
- Easy maintenance
- Scalable design
- Responsive interface
- AI-powered educational support